

Quantification: 543-2

=====

PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image

measure	value
-----	-----
lacuna count	12
interior lacuna count	12
frame-edge lacuna count	0
gap bridges added (detection)	21

Interior lacunae only: mean, median and sample SD (n = 12)

measure	kind	mean	median	sd	units
-----	-----	-----	-----	-----	-----
area_px2	lacuna shape	2027.92	2066	537.141	px^2
major_axis_length_px	lacuna shape	81.5016	86.4661	16.9465	px
minor_axis_length_px	lacuna shape	33.8846	32.3799	6.9212	px
aspect_ratio	lacuna shape	2.4727	2.3986	0.6666	unitless
eccentricity	lacuna shape	0.8942	0.9087	0.0558	unitless
solidity	lacuna shape	0.8887	0.9115	0.0668	unitless
roots_count	headline	7.5833	7.5	2.7455	count
ring_length_r30_px	headline	306.583	309.5	89.1673	px
ring_length_r60_px	reported	966.417	929.5	189.539	px
owned_length_px	ownership-dependent	1687.33	1266.42	1263.6	px
edge_count	ownership-dependent	62.3333	51	44.9188	count
mean_edge_length_px	ownership-dependent	27.4088	28.085	4.2017	px
perimeter_px	normalised	222.101	229.693	45.067	px
ring_area_r30_px2	normalised	8742.17	8905.5	1127.78	px^2
ring_area_r60_px2	normalised	22189.9	23189	2825.36	px^2
in_frame_fraction_r30	normalised	0.9965	1	0.0123	unitless
in_frame_fraction_r60	normalised	0.9814	1	0.0596	unitless
ring_density_r30	normalised	0.0347498	0.0356763	0.00762422	px^-1
ring_density_r60	normalised	0.0434279	0.0446807	0.00529377	px^-1
roots_per_100px_perimeter	normalised	3.3871	3.5267	1.1197	per 100 px
ring_attached_length_r30_px	network v2	232.5	249	89.7689	px
ring_attached_length_r60_px	network v2	572.833	516	228.298	px
ring_length_w_r30_px	network v2	342.996	344.528	104.453	px
ring_length_w_r60_px	network v2	1096.34	1050.65	217.829	px
sholl_crossings_r10	network v2	8.5833	9.5	3.4234	count
sholl_crossings_r20	network v2	12	11.5	3.3845	count
sholl_crossings_r30	network v2	14.5833	14.5	4.5619	count
ring_width_mean_r30_px	width	4.9642	4.9169	0.3688	px

Per field

measure	value	units
-----	-----	-----
analysed_area_px2	1.02424e+06	px^2
total_skeleton_length_px	45785	px
canalicular_length_density_per_px	0.0447014	px^-1
junction_count	681	count
junction_density_per_px2	0.00066488	px^-2
lacuna_count	12	count
lacunae_per_px2	1.172e-05	px^-2

skeleton_component_count	673	count
mean_component_length_px	68.0312	px
median_component_length_px	15	px
field_length_density_w_per_px	0.0506819	px^-1
field_density_without_flagged_per_px	0.0447014	px^-1
field_density_in_roi_per_px		px^-1

Canalicular width, the new measure

measure	value	units
width_mean_px	5.818	px
width_median_px	6	px
width_p10_px	2.8284	px
width_p90_px	8	px
width_pixels_used	45758	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels with real signal under them: bridged pixels, the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2494	90.6112	36.7713	2.4642	0.914	0.9096	0.4535	88.1403	271.181	False
2	3004	86.0791	52.8366	1.6292	0.7895	0.772	0.3181	207.791	919.033	False
3	1698	63.4601	34.3535	1.8473	0.8408	0.955	0.5365	257.425	340.805	False
4	1631	87.5017	30.956	2.8266	0.9353	0.741	0.0248	297.032	38.7695	False
5	1204	55.3793	28.6478	1.9331	0.8558	0.9177	0.1008	394.263	364.613	False
6	2066	81.299	34.8468	2.333	0.9035	0.8619	0.1545	472.247	745.532	False
7	1233	57.0332	27.8479	2.048	0.8727	0.9278	0.052	459.387	484.83	False
8	2066	100.448	26.9826	3.7227	0.9632	0.9018	0.236	550.262	877.61	False
9	2656	108.838	32.5449	3.3442	0.9542	0.8877	-0.0333	642.811	193.327	False
10	2113	91.9731	30.2288	3.0426	0.9444	0.9135	0.0369	728.295	633.991	False
11	2173	86.8531	32.2149	2.6961	0.9287	0.9358	0.0768	787.166	480.067	False
12	1997	68.5432	38.3841	1.7857	0.8285	0.9402	0.1037	859.482	288.042	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	9	366	1178	1615.55	73	22.1308	5.7452
2	12	510	1353	3672.58	146	25.1547	4.916
3	10	316	919	1009.45	35	28.8414	4.683
4	8	305	695	1016.09	52	19.5402	4.5694
5	4	256	826	509.002	18	28.2779	4.5138
6	7	314	898	1516.74	50	30.3348	5.0764
7	2	136	718	352.664	10	35.2664	4.9559
8	10	357	1078	3205.08	103	31.1173	5.3945
9	6	250	940	568.76	22	25.8527	5.3122
10	7	288	1033	826.24	33	25.0376	4.8602
11	9	336	1082	3947.6	134	29.4597	4.9177
12	7	245	877	2008.24	72	27.8922	4.6258

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	244.492	9510	23791	1	0.9843	0.0384858	0.0495145	3.6811
2	295.12	10519	26271	1	1	0.0484837	0.0515017	4.0661
3	172.894	7548	18977	1	1	0.0418654	0.048427	5.7839
4	232.551	8756	18925	0.9575	0.7928	0.0348333	0.0367239	3.4401
5	154.267	7010	17818	1	1	0.0365193	0.0463576	2.5929
6	231.764	9080	22988	1	1	0.0345815	0.0390639	3.0203
7	155.924	7017	19168	1	1	0.0193815	0.0374583	1.2827
8	254.25	9429	23720	1	1	0.0378619	0.0454469	3.9331
9	277.078	10094	25807	1	1	0.0247672	0.0364242	2.1655
10	227.622	8939	23523	1	1	0.0322184	0.0439145	3.0753
11	225.522	8872	23390	1	1	0.037872	0.0462591	3.9907
12	193.723	8132	21901	1	1	0.0301279	0.0400438	3.6134

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	319	821	424.304	1357.71	11	15	18
2	373	954	578.872	1536.78	15	19	23
3	246	472	346.907	1040.72	10	10	12
4	266	560	342.149	776.34	10	13	8
5	158	403	283.693	949.377	6	9	14
6	298	636	356.392	1018.06	10	11	11
7	51	217	147.397	816.896	2	8	9
8	308	871	407.333	1232.16	11	16	16
9	159	360	267.924	1060.59	5	10	12
10	193	429	317.108	1157.1	6	12	16
11	252	720	369.179	1216.43	9	13	21
12	167	431	274.693	993.891	8	8	15

543-2: distributions, pre-validation, px

